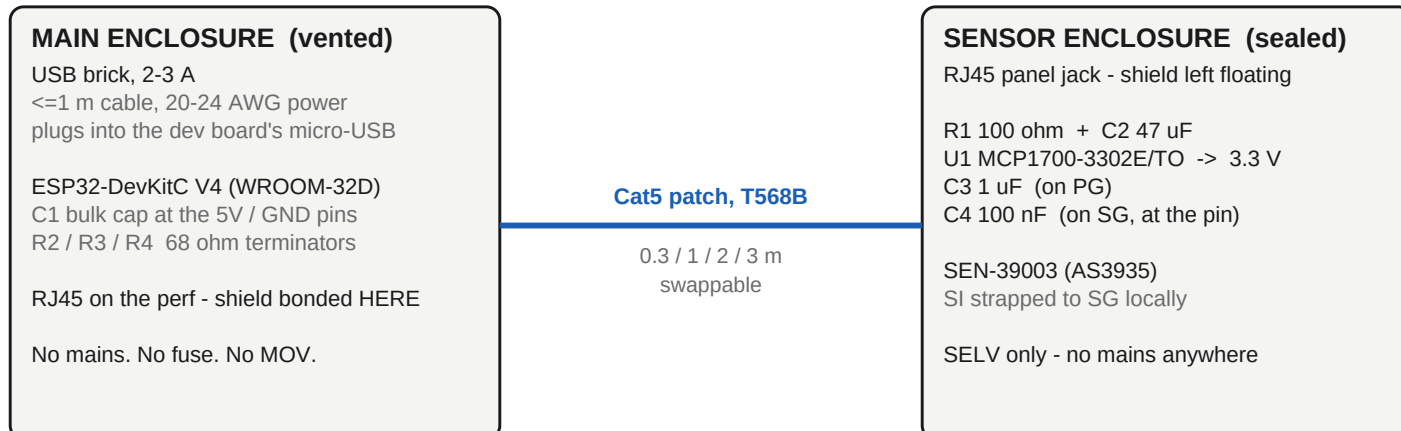


1. System overview

AS3935 lightning detector node - revision 2

Two enclosures. The patch cable length is a variable to be measured, not a fixed design choice - see README section 16.



Why two boxes

Near-field magnetic coupling falls as $1/r^3$, so 5 cm -> 50 cm is roughly a 1000x reduction. Distance is the strongest lever available and nothing else is close. The interference source was never identified (README 11.3), so the separation is left adjustable on purpose: survey at each cable length and read the curve. A flat curve eliminates the ESP32 and the supply properly for the first time.

Do not plug this into a network switch

The RJ45 carries 5 V and SPI, not Ethernet. A live PoE port puts 48 V on those lines and destroys the ESP32 and the sensor. Label both ends.

Signal assignment

T568B at both ends. Pairs are (1,2) (3,6) (4,5) (7,8). Seven signals plus one spare, and the spare is spent on a second ground paired with SCLK so the fastest edge gets its own return. SI is not carried - it straps to ground inside the sensor box.

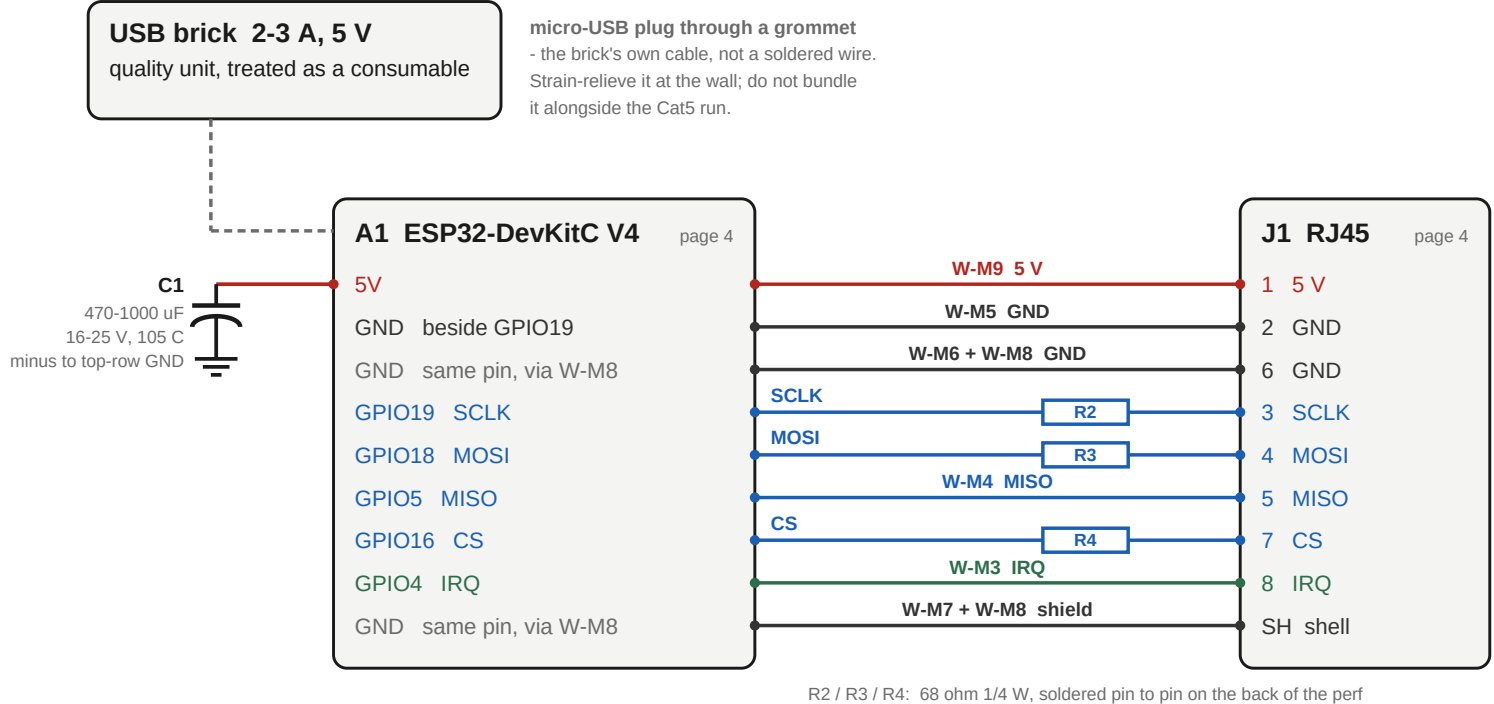
The jack has a ninth pin

The panel breakouts are shielded jacks on a 9-way 0.1 in header: 1-8 plus SH, the metal shell. SH is bonded to ground at the MAIN board only and left floating at the sensor. One end only - bonding both would wrap a ground loop around the whole run if a shielded patch cable were ever fitted. Pinout and orientation: page 4.

2. Main enclosure

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USB supply, ESP32, series terminators and the RJ45 jack. Every line is one physical wire.



R2 / R3 / R4 - 68 ohm series terminators. Fit them; do not leave links.

Only the three lines the ESP32 DRIVES get one: SCLK, MOSI, CS. MISO and IRQ are driven from the far end, so damping them at THIS end would do nothing at all.

What makes the cable electrically long is the EDGE rate, not the clock rate: the ESP32's edges stay a few ns however slowly you clock it, and Cat5 runs ~5 ns/m, so the round trip is ~10 ns at 1 m and ~30 ns at 3 m. data_rate: 200kHz removes the TIMING consequence of the ringing, not the ringing itself.

Unterminated, the ~2.5 V launched into a ~100 ohm line doubles at the AS3935's high-impedance input and its ESD clamps dump the excess INTO the sensor's local 3.3 V rail, on every clock edge, in the one place this whole design exists to keep quiet.

68 ohm makes $Z_{out} + R$ match the line, so the returning wave is absorbed. Cost at 200 kHz: a ~10 ns rise against a 5 us bit period. Clean match for SCLK, whose pair partner is its own return (pins 3 and 6); MOSI and CS return through whatever ground is nearest, so for those 68 ohm is an approximation.

C1 is the reason this board exists

It is the reservoir for the 300-500 mA WiFi bursts the USB cable's resistance cannot supply fast enough, so it must sit AT the 5V and GND pins. The DevKitC has five pins between 5V and its nearest ground, so the loop is ~27 mm however you arrange it - a property of the dev board, not of the layout. Watch the polarity.

The USB cable is a circuit element, not an accessory

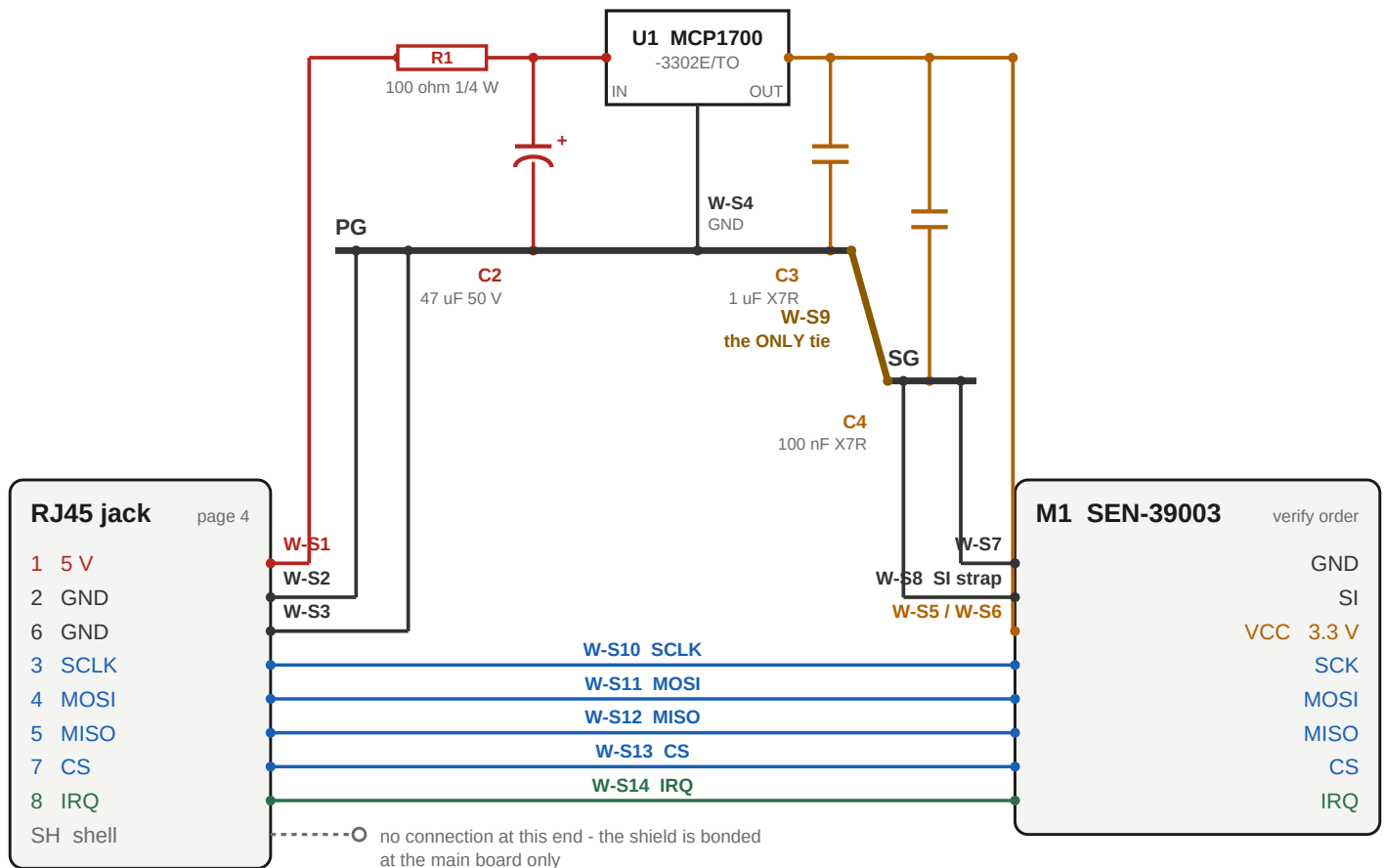
28 AWG conductors are ~0.21 ohm/m. Over 2 m, counting the ground return, that is ~0.84 ohm: a 500 mA burst drops ~0.42 V, so 5.0 V arrives as 4.58 V, and the onboard AMS1117 - which needs over a volt of headroom - browns out. The IRM-02-5 failure by a different route.

Use <=1 m with 20-24 AWG power conductors, and MEASURE at the 5V pin under WiFi load. Keep the USB cable away from the Cat5 run; do not bundle them parallel.

3. Sensor enclosure

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Every part of the local rail, drawn. 5 V arrives on the cable; 3.3 V is made here, centimetres from the pins.



Two grounds, and exactly one tie between them

PG carries the cable's ground return, C2, and the regulator's own reference. SG carries only the sensor's GND pin, C4 and the SI strap. They meet at W-S9 and nowhere else. C3 belongs to the regulator (stability) so it lands on PG; C4 belongs to the sensor (decoupling) so it lands on SG. Swapping them defeats the split as surely as a second tie. A second bridge wraps a ground loop around the LDO and C4 stops being local.

Why a linear regulator, and why out here

A switching regulator would put a 100 kHz - 1 MHz noise source centimetres from a 500 kHz magnetic antenna: the worst possible place for one. A linear regulator has no switching node, and at under 1 mA the wasted heat is nothing. The E suffix is the -40/+125 C grade, required for a ~52 C attic. C3 is REQUIRED for MCP1700 stability, not optional. LDO rejection is largely gone by 500 kHz, so R1/C2 and C3/C4 are complementary to it, not redundant with it. Neither the regulator nor the passives suffices alone.

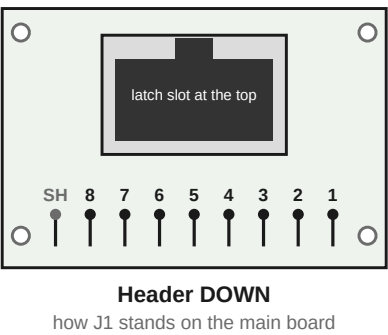
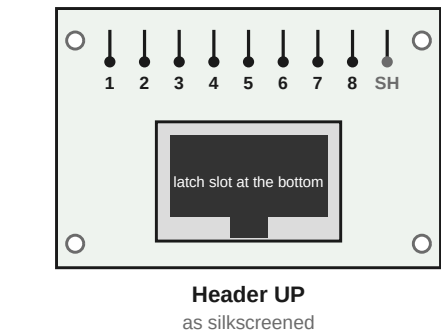
Headroom check

5 V leaves the main board from the dev board's 5V pin, which on some DevKitC boards sits behind a Schottky and reads ~0.3 V low. R1 drops another ~0.1 V at the sensor's sub-1 mA draw. Worst case the MCP1700 still sees ~4.5 V for a 3.3 V output - enormous headroom. Measure it anyway at bring-up: 3.3 V at the sensor VCC pin, at the far end (README 12).

4. Connector and board pinouts

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The two parts this design is most likely to be wired backwards to. Wire to the printed numbers, never to a position in a photograph.



Both drawings are the SAME face - the jack side, which is also the silkscreen side, turned half a turn. Which order you see depends on which way up the header is, NOT on which side you look from.

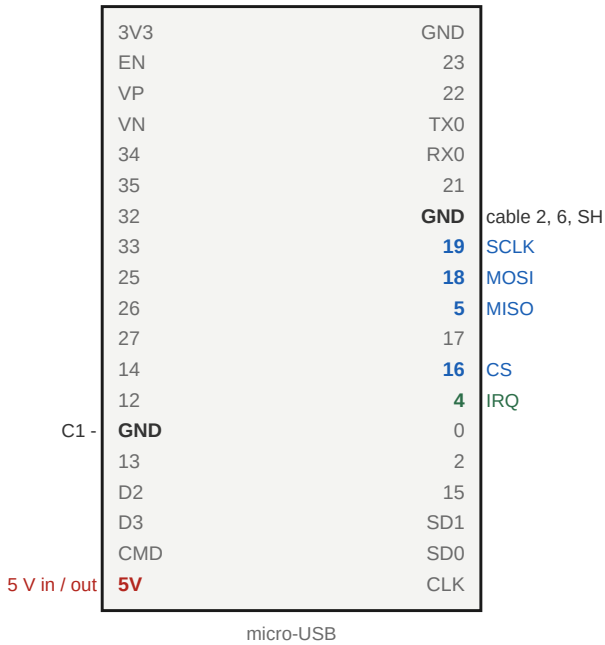
On the main board J1 stands on edge, header down, jack facing off the board edge: the right-hand view. Hence SH at the USB end.

Drill from the part, not a drawing.

Pin	Signal	T568B colour	Note
1	5 V	white/orange	
2	GND	orange	
3	SCLK	white/green	
4	MOSI	blue	
5	MISO	white/blue	
6	GND	green	
7	CS	white/brown	
8	IRQ	brown	
SH	shield	(jack shell)	bonded to GND at the main board only

The jack is top-entry: the cable goes in PERPENDICULAR to the breakout PCB. MAIN board: J1's right-angle header solders straight into the perf, so it stands on edge with the jack out through the enclosure wall - and the wall, not the nine header joints, must take the force of plugging a cable in. SENSOR board: the breakout lies flat against the inside of the wall behind a ~17 x 16.5 mm cutout, with a Cat5e pigtail to the perf.

A1 ESP32-DevKitC V4 / ESP32-WROOM-32D - 38 pins, USB at the bottom



Measured off the vendor pinout: 19 pins per row, rows 1.0 in (25.4 mm) apart - exactly 10 holes on 0.1 in perf.

These are NOT the default VSPI pins. On the main board each resistor sits pin to pin between an ESP32 pin and the jack pin directly below it, and neither pin order can move. So the GPIOs were chosen to line up with the jack: SCLK 19, MOSI 18, MISO 5, CS 16, IRQ 4. The ESP32 routes SPI to any pin; at these speeds the GPIO matrix costs nothing. Do not 'fix' them back to the defaults.

(The vendor diagram mislabels left pin 10: the silkscreen reads 26, not 23.)

5V's nearest ground is five pins away.

5. Wire schedule

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Reference numbers match pages 2 and 3, and the hole coordinates in as3935-protoboard-layout.pdf.

Ref	From	To	AWG	Colour	Note
W-M1	C1 +	ESP32 5V pin	22 solid	red	as short as physically possible
W-M2	C1 -	ESP32 GND, top row	22 solid	black	stripe lead is minus
W-M3	ESP32 GPIO4 IRQ	J1 pin 8	24 solid	green	pin to pin; no resistor
W-M4	ESP32 GPIO5 MISO	J1 pin 5	24 solid	blue	pin to pin; no resistor
W-M5	ESP32 GND beside 19	J1 pin 2	24 solid	black	solder to (14,15) in passing
W-M6	J1 pin 6	W-M8 tap (10,15)	24 solid	black	the SCLK pair's return
W-M7	J1 pin SH	W-M8 tap (7,15)	24 solid	black	shield bonded at THIS end only
W-M8	(7,15)	(14,15)	24 solid	black	COMPONENT SIDE; tap (10,15)
W-M9	C1 +	J1 pin 1	24 solid	red	round the header end, col 21
R2	ESP32 GPIO19 SCLK	J1 pin 3	68 ohm	-	pin to pin on the back
R3	ESP32 GPIO18 MOSI	J1 pin 4	68 ohm	-	pin to pin on the back
R4	ESP32 GPIO16 CS	J1 pin 7	68 ohm	-	pin to pin on the back
W-S1	RJ45 pin 1	R1 100 ohm	24 solid	wh/orange	start of the local rail
W-S2	RJ45 pin 2	PG rail	24 solid	orange	
W-S3	RJ45 pin 6	PG rail	24 solid	green	the SCLK pair's return
W-S4	U1 GND pin	PG rail	24 solid	black	
W-S5	U1 OUT 3.3 V	C4 + / VCC node	24 solid	red	
W-S6	C4 + node	sensor VCC	24 solid	red	one hole - do not lengthen
W-S7	sensor GND	SG rail	24 solid	black	
W-S8	sensor SI	SG rail	24 solid	black	grounds SI: selects SPI, not I2C
W-S9	SG rail	PG rail	22 solid	black	THE ONLY TIE - mark it
W-S10	RJ45 pin 3	sensor SCK	24 solid	wh/green	
W-S11	RJ45 pin 4	sensor MOSI	24 solid	blue	
W-S12	RJ45 pin 5	sensor MISO	24 solid	wh/blue	
W-S13	RJ45 pin 7	sensor CS	24 solid	wh/brown	
W-S14	RJ45 pin 8	sensor IRQ	24 solid	brown	
-	RJ45 pin SH (sensor)	nothing	-	-	left floating: single-point shield

Nothing here is carrying any current worth the name

Past the RJ45 the whole sensor rail draws under 1 mA. The longest run on either board is about 90 mm; even 30 AWG would add 30 milliohms, or 15 microvolts at 500 uA. R1 drops 0.1 V on purpose - six thousand times more. Gauge is electrically irrelevant on both boards. It is chosen on mechanical grounds alone, and page 6 says why.

The one place gauge ever mattered in this project is the USB cable (README 7.4), and that is a cable you buy, not one you build.

Colour is documentation, so keep it

The sensor board's links carry T568B colours because its pigtail is cut from Cat5: the conductor colours then match the pin numbers on the breakout and the cable between them, end to end. The main board has no pigtail - J1 solders straight in - so its wires use plain red 5 V, black ground, blue SPI and green IRQ, as do the sensor wires that are not part of the cable run (W-S4, W-S7, W-S8, W-S9).

6. Wire specification, and bring-up

AS3935 lightning detector node - revision 2

What to buy, what not to use, and the order to switch things on in.

What each job actually needs

Where	What	Length	Why
Buses on the sensor board	bare SOLID tinned copper, 20-22 AWG	~1 m	must lie straight across the pads
Links on both boards	insulated SOLID, 24 AWG (26 also fine)	~2 m	must enter a 1 mm hole unaided
Pigtail, sensor board	solid Cat5e offcut, 8 cores	~15 cm	main board has none: J1 solders in
USB brick to ESP32	none - it plugs into the micro-USB	-	grommet and strain relief only
Mains variant (README 7.3)	stranded silicone, 18-20 AWG	-	not built in rev 2

Shopping list

Item	Specification	Qty	Note
Bus wire	bare solid tinned copper, 22 AWG, ~0.64 mm	1 spool	sold as 'buss bar wire' / 'jumper wire'
Hookup wire	solid core, 24 AWG, PVC or PTFE, 6 colours	1 kit	PTFE if you can: it will not melt back
Cat5e offcut	SOLID conductor cable, ~1 m	1	riser/in-wall stock, NOT a patch cable
Female header	0.1 in, 1x40 breakaway	2	cut to 1x19 for the ESP32
Male header	0.1 in, 1x40 breakaway	1	8 pins for the SEN-39003, if not supplied
Resistors	68 ohm 1/4 W metal film	3	R2 / R3 / R4 - see page 2

Why not the silicone stranded already on hand

16 / 18 / 20 / 24 AWG stranded silicone is excellent wire and the wrong wire for this. Stranded will not enter a 0.1 in hole without being tinned first, and a tinned end is a solid end with worse geometry. Silicone insulation is thick and soft, so at 2.54 mm pitch it crowds neighbouring holes and will not hold a route. And a bus has to be a straight bare bar soldered to eight pads in a row: stranded cannot be made straight. Keep it for the mains variant and for anything that has to flex.

Why solid Cat5e for the sensor pigtail

Eight solid 24 AWG conductors in one jacket, already coloured to T568B, so the pigtail documents itself against the pin table on page 4. Do not cut up one of the patch cables bought for the distance sweep - those are the experiment. Buy a metre of in-wall stock, or salvage a dead cable, and check it is SOLID: patch cable is stranded. Its insulation is usually HDPE and shrinks back fast under an iron. Strip generously, tin quickly, do not dwell. Leave a service loop and cable-tie both ends to the board. SH has no conductor in the cable, so it is a ninth wire: any offcut will do.

Not 30 AWG wire-wrap, tempting as it is

Kynar wire-wrap is the classic perfboard wire and it is genuinely nicer to route. It is also fragile, and README 11.3 is this project's warning about builds that move: the old breadboard's noise floor fell by two thirds the moment it was handled. Rigidity is a pass/fail measurement here, so spend the extra bulk on 24 AWG solid.

Bring-up, in order

1. Assemble both boxes; connect with the SHORTEST patch cable.
2. Measure 5 V at the ESP32 5V pin DURING WiFi activity, not at idle. Want >4.7 V.
3. Measure 3.3 V at the sensor VCC pin, at the far end, after the LDO.
4. Verify tuning capacitance OVER SERIAL - the one check WiFi cannot do (README 12.1).
5. Confirm the sensor responds to the SEN-39002 emulator. Expect disturbers, not lightning.
6. Platform-validity test: survey, handle the build, survey again. Rates must not move.
7. Only then mount, and run the distance sweep at 0.3 / 1 / 2 / 3 m.